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substrateminer: A Python package to investigate protein substrate repertoires

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Abstract

`Substrateminer` is a Python package that provides a suite of investigative tools to develop protein substrate repertoires for a given protease of interest. `Substrateminer` provides tools for analysing protease cleavage consensus and visualising the conservation and strength of consensus. `Substrateminer` is a complete suite that enables potential substrate identification for a protease of interest in a target organism and extended functions allow for further discovery of associated biological and pathological functions of detected substrates.

Introduction

Enzyme hydrolysis is a fundamental biochemical process in biology, and underpins many biological processes in nature from food digestion in humans (Campbell & Reece, 2005) to renewable energy production with cellulosic ethanol (Guo et al., 2023; Wright, 1988). Proteases are a class of enzymes that carry out such hydrolytic functions in biology by cleaving peptide bonds. In human physiology, a wide variety of critically important biophysiological processes depend on well-regulated protease functions, including protein degradation, cell signalling, and immune response. (Bond, 2019; Kammerl & Meiners, 2016; López-Otín & Bond, 2008) Consequently, the inhibition of target proteases has long been one intervention strategy applied clinically to treat diverse pathophysiological conditions including viral infections like HIV, hepatitis-C; metabolic dysfunctions like type 2 diabetes (Scott, 2017); and increasingly, cancers (Manasanch & Orlowski, 2017). In the wake of the COVID-19 pandemic and increased respiratory diseases, proteases have emerged as a major therapeutic target for curbing fatality due to viral infections.(Borges et al., 2024; Papaneophytou, 2024; Sojka et al., 2021)

Nevertheless, the protease network in many organisms especially mammals including humans is complex and heavily-entangled, and the regulatory scope for many target proteases is still not well-understood. The ability to gather and identify novel protease substrates is an important step towards a better understanding of the biological functions of proteases and accelerate future developments of new protease inhibitors for downstream therapeutic applications. Here, the introduction of `substrateminer` aims to provide a comprehensive suite of tools to streamline the investigation and develop protein substrate repertoires for a given protease of interest based on a consensus algorithm.

Methods

`Substrateminer` consists of three main categories of functions (Figure 1), namely consensus investigation (i.e., with modules msa and consensus), substrate mining (i.e., with module miner) and substrate biopathological pathway searching (i.e., with module pathfinder).

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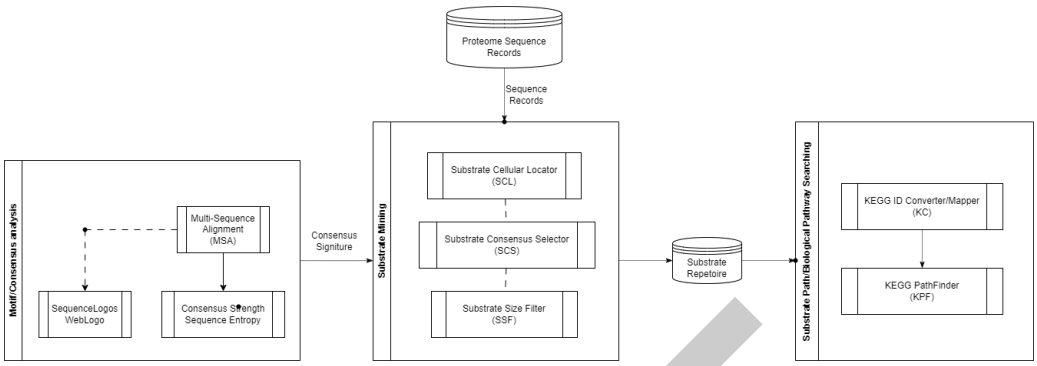

Figure 1: Substrateminer workflow

A typical workflow for adopting `substrateminer` to investigate protein substrate repertoires for a given protease often follows three main steps:

Consensus investigation

For a target protease with a few known cleaving substrates, finding a cleavage consensus is often beneficial. `Substrateminer` provides access to a mechanism to visualise the cleavage consensus motif and calculate the strength of conservation at each site along the motif. The strength of conservation is calculated on entropy and `substrateminer` is also packaged with auxiliary tools to provide multi-sequence alignment (MSA) access to help prepare the input data for consensus investigation. Consensus visualisation is based on the implementation of weblogo (Crooks et al., 2004). (Figure 2)

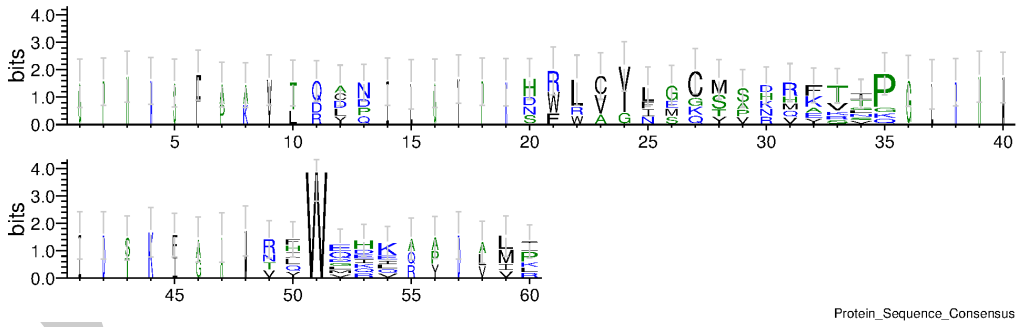

Figure 2: Consensus investigation - Visulisation Example

Substrate mining

Upon the identification of a cleavage consensus, `substrateminer` provides a mechanism to mine potential substrates based on three filter strategies:

1. *Cellular location*: filter potential substrates based on the cellular localisation of the intended target. For example, if the target protease is a secreted protease, the substrates mined should be extracellular proteins.
2. *Target Size*: filter potential substrates based on the size of the intended target. For example, if the target substrate is a small protease, the substrates mined should be small proteins or peptides to a size of determined value.
3. *Consensus*: filter potential substrates based on the conservation of the cleavage consensus. At the current release, both endopeptidase and exopeptidase cleavage consensus are supported.

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Substrate pathobiological pathway searching

To further investigate the pathobiological relevance of novel substrates, `substrateminer` implemented a search strategy to retrieve known biological processes and disease associations based on the *de facto* standard KEGG database (Kanehisa, 2019; Kanehisa & Goto, 2000). In the present release, the `pathfinder` module in the `substrateminer` provides three main functions namely KEGG accession conversion, biological processes tracing and human disease mapping, where KEGG accession conversion enalbes a gateway to KEGG databases where further information including enzyme nomenclature and diseases-related network databases can be easily accessible. A more comprehensive implementation of KEGG access is scheduled for future releases.

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